

Evaluation of chromosomal aberrations, micronuclei, and sister chromatid exchanges in hospital workers chronically exposed to ionizing radiation.

Cytogenetic analysis was performed in peripheral blood lymphocytes from hospital workers chronically exposed to ionizing radiation in comparison to matched non-exposed individuals. The accumulated absorbed doses calculated for the radiation workers ranged from 9.5 to 209.4 mSv. The endpoints used were chromosomal aberrations (CA), micronuclei (MN), and sister chromatid exchanges (SCE). The frequencies of CA/100 cells observed for the exposed group were significantly ($P=0.018$) higher than in the control group: 3.2 and 2.6, respectively. Similarly, the mean numbers of SCE per cell were statistically higher ($P=0.025$) in the exposed group (6.2) in comparison with the control group (5.8). In the case of micronuclei analysis, no significant ($P=0.06$) difference between both groups was found, but these data should be cautiously interpreted since an increase in the frequencies of MN was found for radiation workers (3.0 MN/100 cells), compared to the control group (2.6 MN/100 cells) and this increase occur in parallel to CA and SCE frequencies. The difference between the results could be explained by the nature of CA and MN generation. The increased frequencies of CA and SCE in radiation workers indicate the cumulative effect of low-level chronic exposure to ionizing radiation, and the relevance of conducting cytogenetic analysis in parallel to physical dosimetry in the working place.

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